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FIRST SEMESTER B. Tech EXAMINATION, JANUARY- 2024

APPLIED CHEMISTRY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO	Marks
		PART-A	**5 X 2=10**		
1.	**a.**	Define Standard reduction potential (E^o)?	L-1	CO1	**02**
	b.	Explain internal energy.	L-1	CO2	**02**
	c.	What is effective nuclear charge?	L-2	CO3	**02**
	d.	Describe optical activity.	L-3	CO4	**02**
	e.	What are nanomaterials?	L-1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Discuss the construction and working principle of standard hydrogen electrode.	L-2	CO1	**05**
		OR			
	b.	Differentiate between galvanic cell and electrolytic cell.	L-2	CO1	**05**
3.	**a.**	Articulate how different variables impact on the equilibrium constant.	L-3	CO2	**05**
		OR			
	b.	List out the difference between reaction quotient and equilibrium constant.	L-2	CO2	**05**
4.	**a.**	Illustrate the atomic size and electronegativity along the periods and groups.	L-3	CO3	**05**
		OR			
	b.	Draw the molecular orbital diagram of O_2 and calculate the bond order.	L-3	CO3	**05**
5.	a.	What is structural and geometrical isomerism? Give example.	L-2	CO4	**05**
		OR			
	b.	Define substitution reaction and explain SN_2 reaction mechanism.	L-1	CO4	**05**
6.	**a.**	Describe the working principle of XRD instrument	L-3	CO5	**05**
		OR			
	b.	Summarize the futuristic applications of nanomaterials.	L-2	CO5	**05**

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7.	**a.**	Summarize the differential metal and differential aeration corrosion.	L-2	CO5	**05**
		OR			
	b.	Extract the bond order and magnetic behavior of N_2 molecule using molecular orbital diagram.	L-2	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Derive Nernst's equation for the single electrode reaction.	L-2	CO1	**10**
		OR			
	b.	Discuss the factors affecting on rate of corrosion.	L-2	CO1	**10**
9.	**a.**	Illustrate the characteristics of Ellingham diagram with neat labeled diagram.	L-1	CO2	**10**
		OR			
	b.	Summarize the contact procedure for synthesizing H_2SO_4 and explain Le-principle Chatelier's in its synthesis.	L-3	CO2	**10**
10.	**a.**	Explain briefly about classification of elements in modern periodic table.	L-2	CO3	**10**
		OR			
	b.	Extract the bond order, magnetic behavior and molecular orbital configuration of CO using molecular orbital diagram.	L-3	CO3	**10**
11.	**a.**	What is conformational analysis? Discuss the conformational analysis of ethane.	L-3	CO4	**10**
		OR			
	b.	How do you prepare Paracetamol and Acetylsalicylic acid (Aspirin).Write the reaction?	L-3	CO4	**10**
12.	**a.**	Explain the synthesis of nanoparticles by using sol- gel method.	L-2	CO5	**10**
		OR			
	b.	Summarize the principle, working mechanism and applications of Fourier transform infrared spectroscopy.	L-3	CO5	**10**
13.	**a.**	What is crystal field splitting? Explain octahedral splitting in complexes.	L2	CO3	**10**
		OR			
	b.	Define the term conformational analysis and discuss the conformational analysis of cyclohexane molecule.	L-1	CO1	**10**